Statistical and Computational Methods for Network Data

# HUMAN CONNECTOME ANALYSIS

Giorgia Cappellato - Matteo Cristina - Elisa Quadrini

# 1. INTRODUCTION

**1. Dataset & Network Architecture**

- The Connectome: A complete wiring diagram of the healthy human brain, merging Diffusion MRI (dMRI) scans of 477 individuals across 1,015 distinct regions.
- Nodes (Grey Matter): The brain's processing regions, located primarily in the cortex.
- Edges (White Matter): Long nerve fibers (axons) acting as communication cables.

**2. Project Origins & Objective**

- Origins: Raw data collected by the Human Connectome Project (HCP) and assembled by the PIT Bioinformatics Group at Eötvös Loránd University in Budapest.
- Objective: Apply network analysis to uncover the structural patterns of how the brain wires itself to share information.

**3. Key Network Edge Attributes**

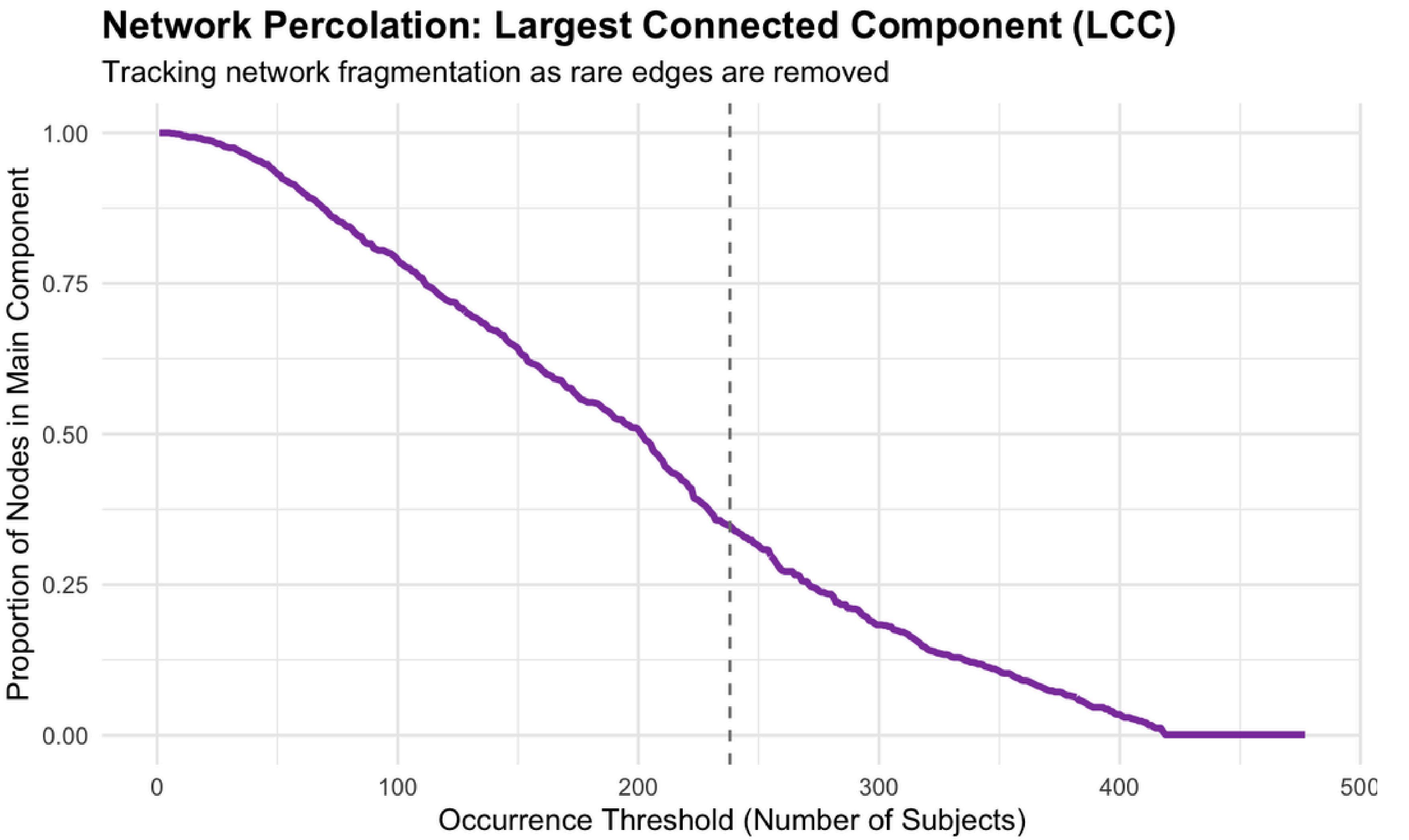
- *fiber_count_mean*: Evaluates the physical robustness of a connection by counting the average size of nerve fibers linking two regions.
- *electrical_connectivity_median*: Estimates the functional efficiency of the pathway, measuring how well electrical signals travel down those fibers.

# 2. FILTERING BIOLOGICAL NOISE

Aggregating tractography across 477 subjects introduces systemic false positives and highly individualistic peripheral wiring, sculpted by genetics and environment.

As we remove rare edges, the network fragments steadily. With a fixed 50% subject consensus threshold, only ~35% of nodes survive.

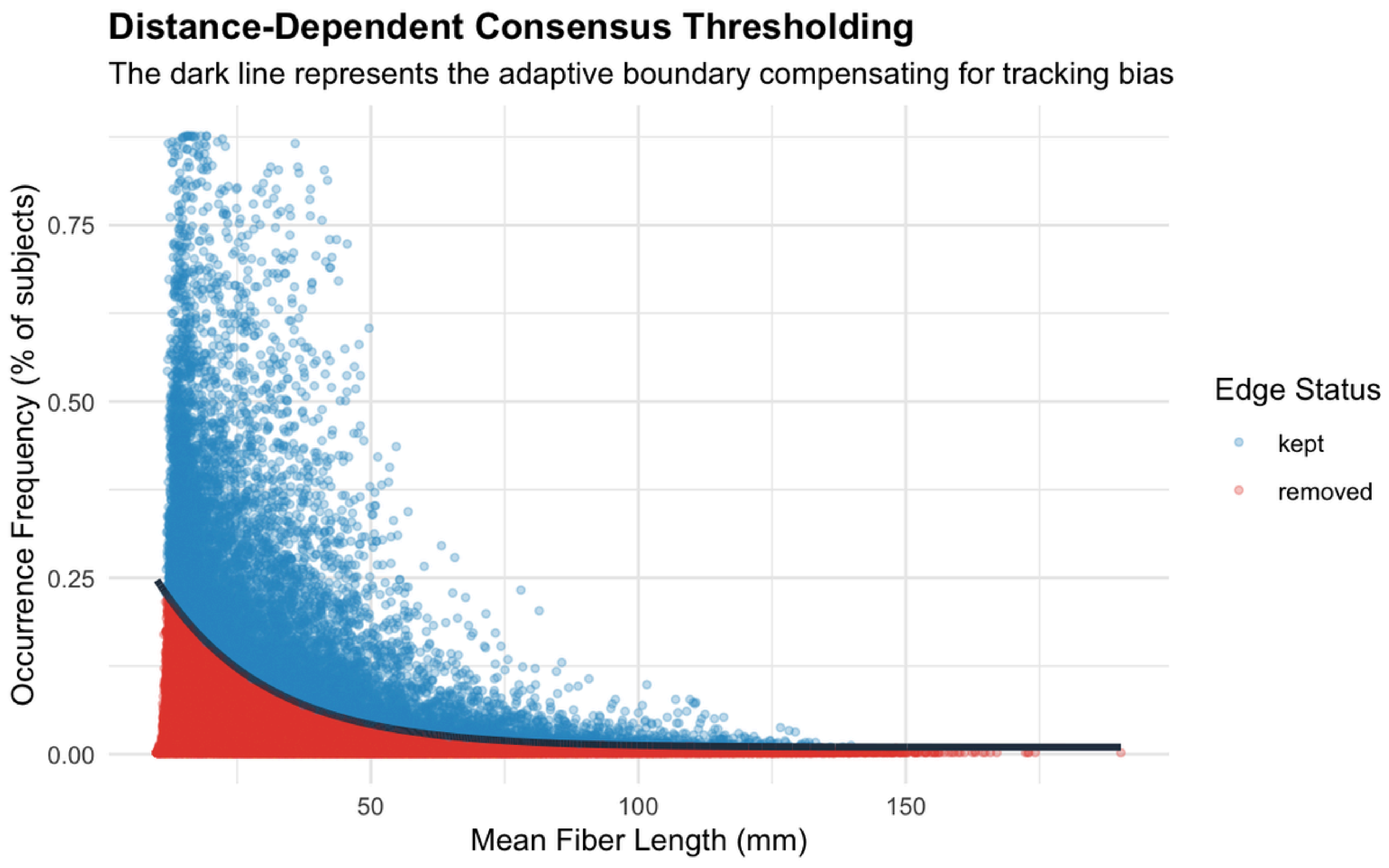
But this traditional rigid thresholds assume false positives occur equally everywhere.



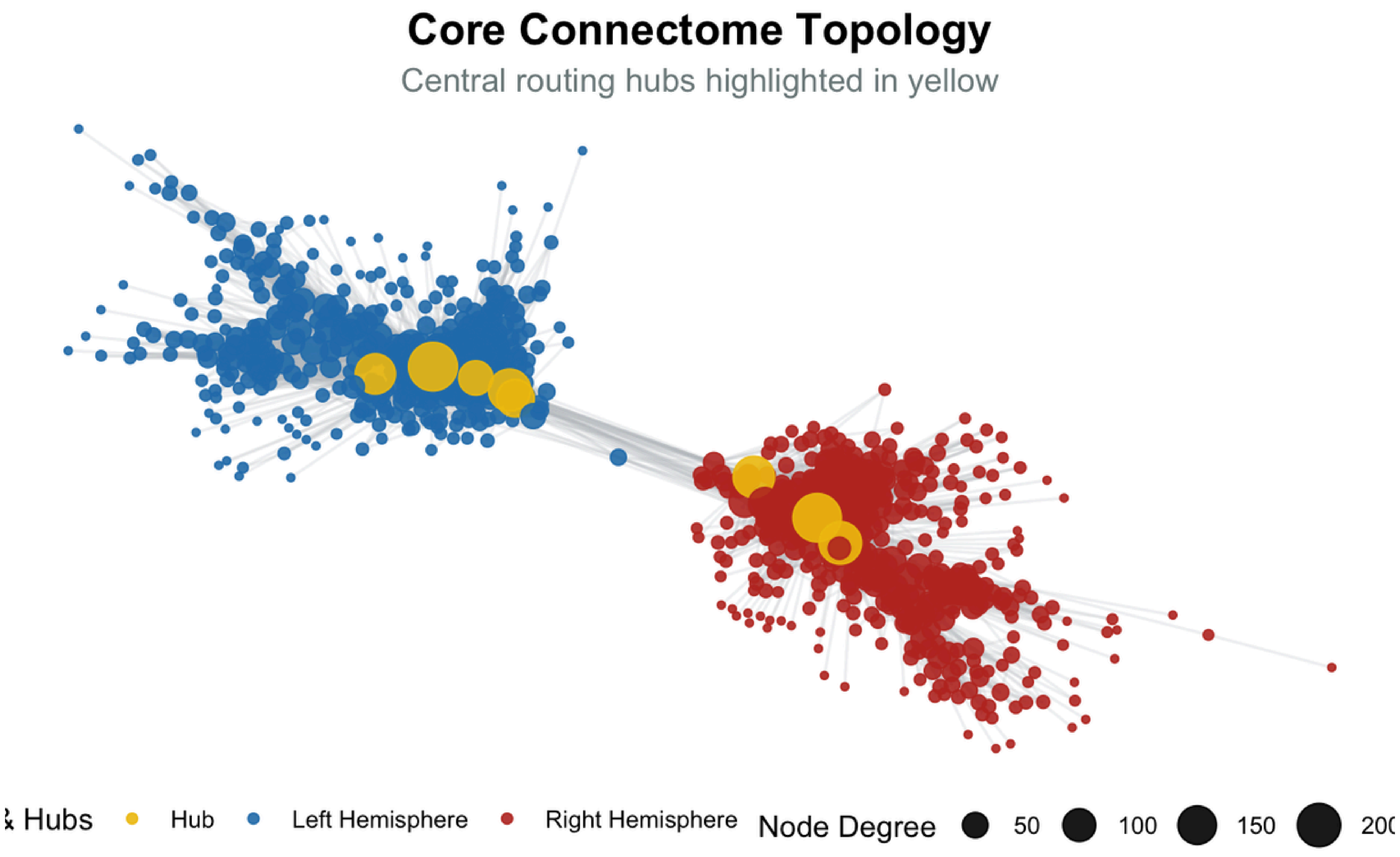
# 2.1 DISTANCE-DEPENDENT THRESHOLDING

In reality, dMRI technology inherently struggles to trace very long nerve fibers, especially those bridging the two hemispheres.

To preserve long range structural connections without keeping short range noise, we applied a dynamically decaying exponential threshold based on physical fiber length.



# 3. SUBGRAPH VISUALIZATION




The denoised connectome is visualized using the Fruchterman-Reingold layout.

Even without explicit spatial coordinates, this physical simulation autonomously segregates the network into distinct left (blue) and right (red) hemispheres.

To emphasize network importance, node sizes are scaled proportionally to degree centrality. The brain's central routing hubs illustrate how these routing centres form the dense core of each hemisphere.

# 4. DEGREE DISTRIBUTION

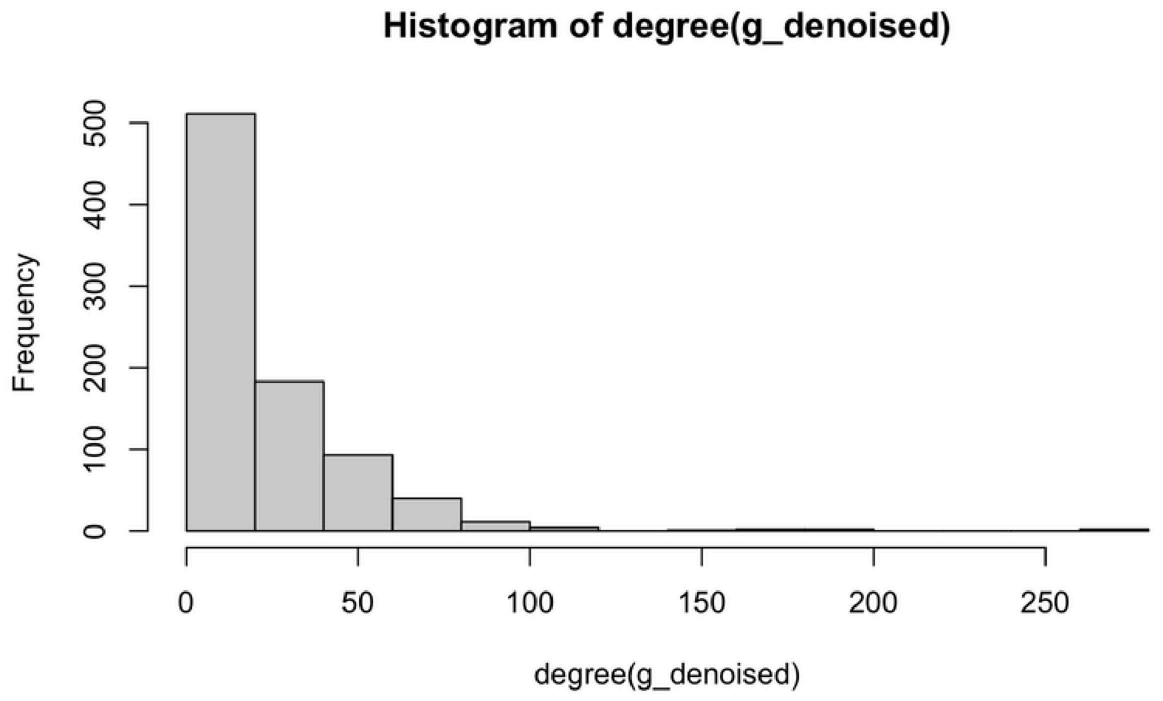
The network exhibits a strongly right-skewed degree distribution, suggesting an heterogeneous behavior.

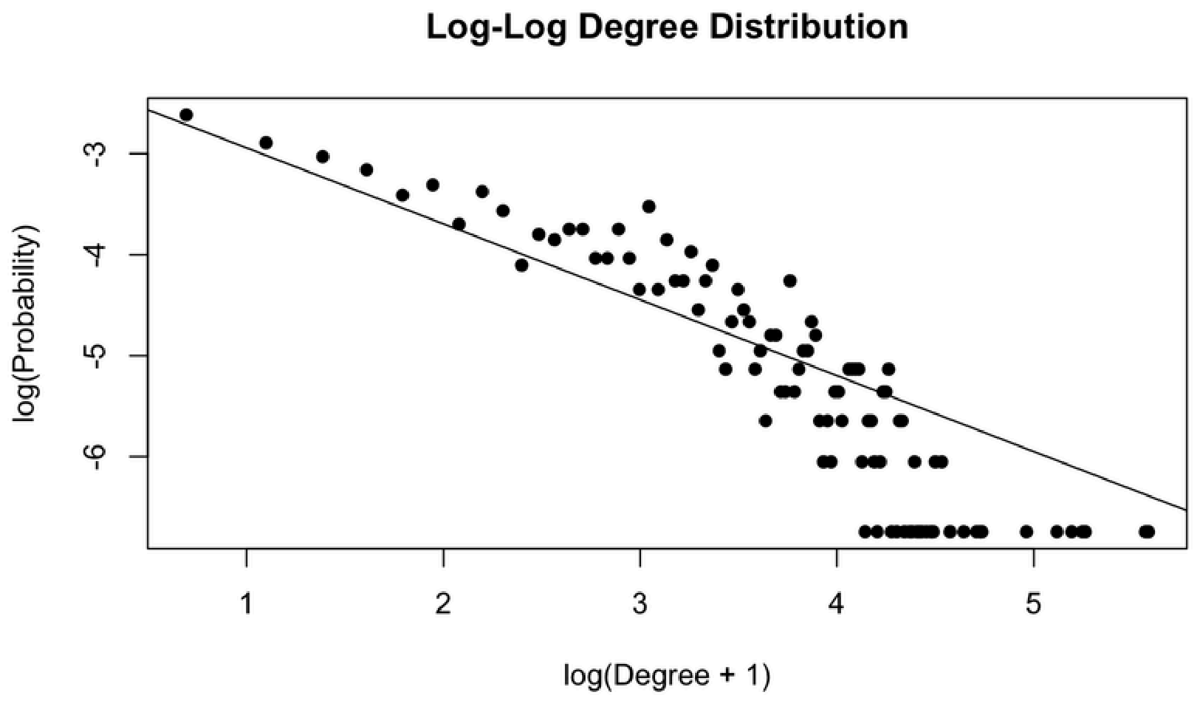
Testing for Scale-Free Architecture: we used MLE to strictly isolate and evaluate the extreme tail of the network, finding the optimal threshold via KS minimization.

Key Statistical Findings

- *xmin*=57: Power-law behavior begins only for hubs with 57 or more connections.
- *α*=4.29: The scaling exponent shows a steep decay in the probability of finding massive hubs.
- *p*=0.408: A statistically plausible fit for a power-law model in the network tail.

The Biological Constraint: The human brain is constrained by the skull volume and the ATP consumption. Therefore, biology dictates a strict limit on their maximum size, forcing extreme hubs to decay much faster (*α*=4.29).



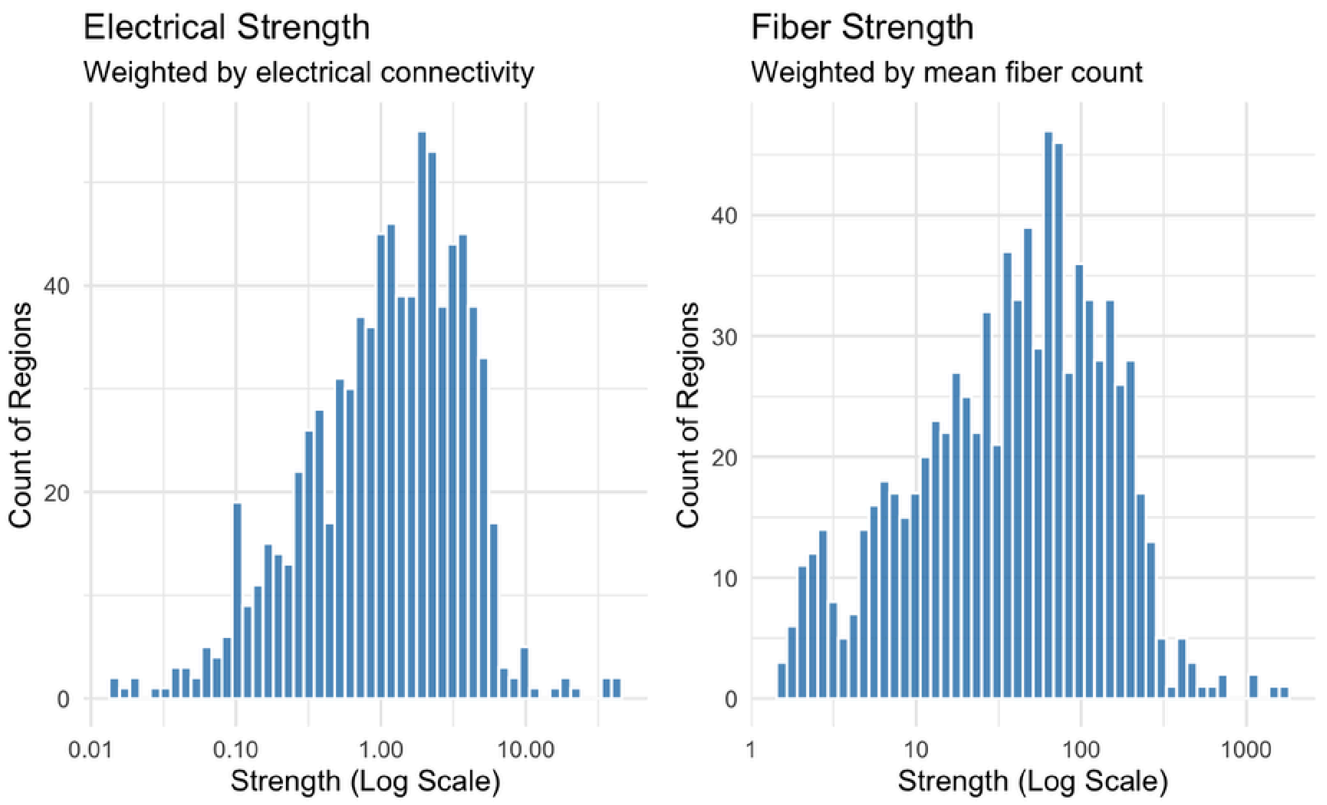



# 4.1 MAPPING STRUCTURAL AND ELECTRICAL HUBS

While most of the brain is moderately wired, a few elite hubs command massive physical and electrical bandwidth.

After mapping the hubs to their anatomical brain regions, we analyzed their biological efficiency by computing an electrical-to-fiber strength ratio. Findings:

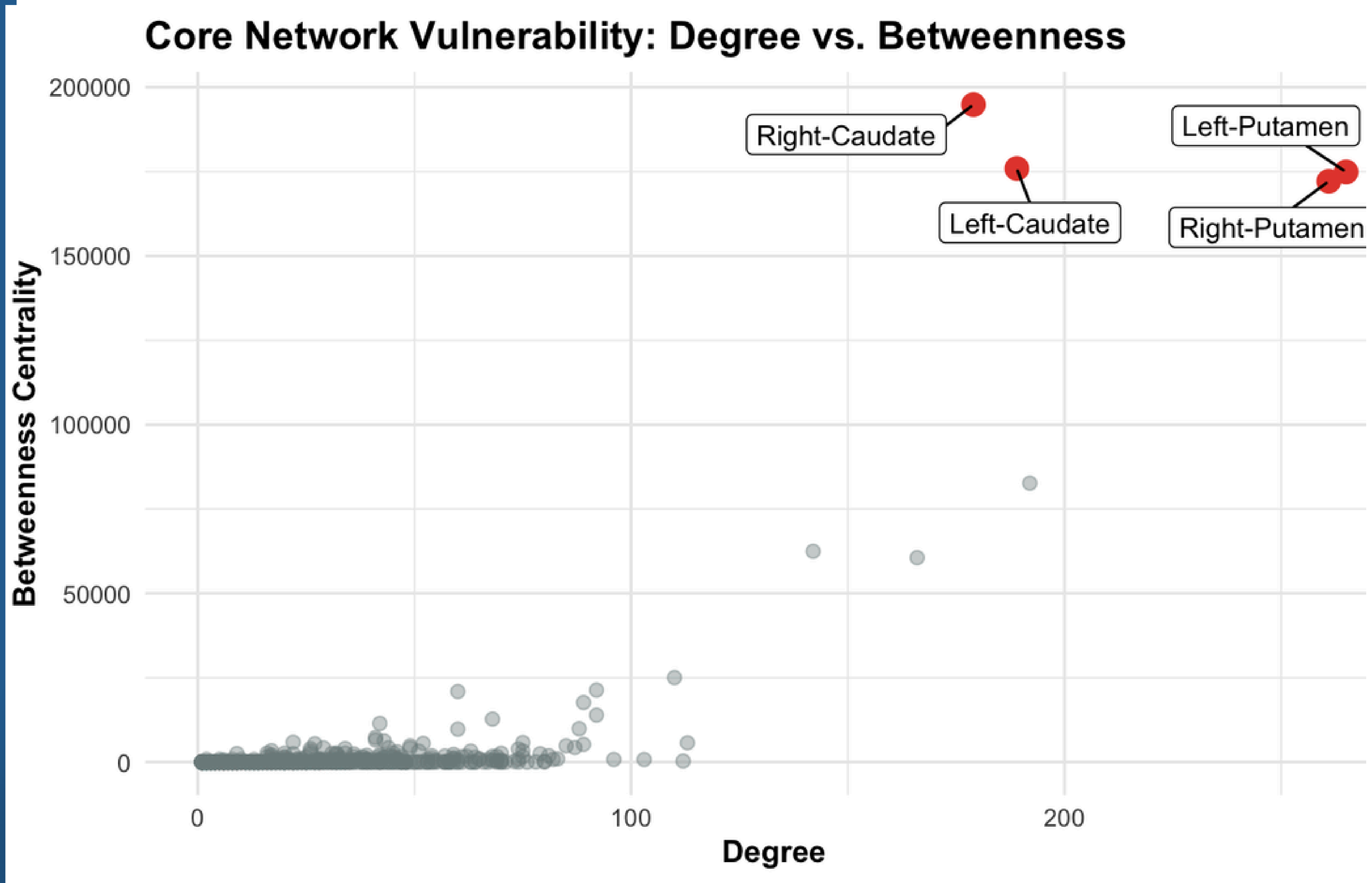
- The Subcortical Core: 8 of the top 10 hubs are subcortical structures. These centers control sensory routing, motor commands, and memory.
- Biological Symmetry: For almost every right-hemisphere hub, its left counterpart ranks immediately adjacent. This mirroring validates the accuracy of our data cleaning.
- Two Distinct Strategies: The Putamen acts as a massive highway, relying on a high volume of raw physical fibers to move traffic. The Caudate uses 30% fewer connections, but highly optimizes each pathway for maximum electrical signal transmission.

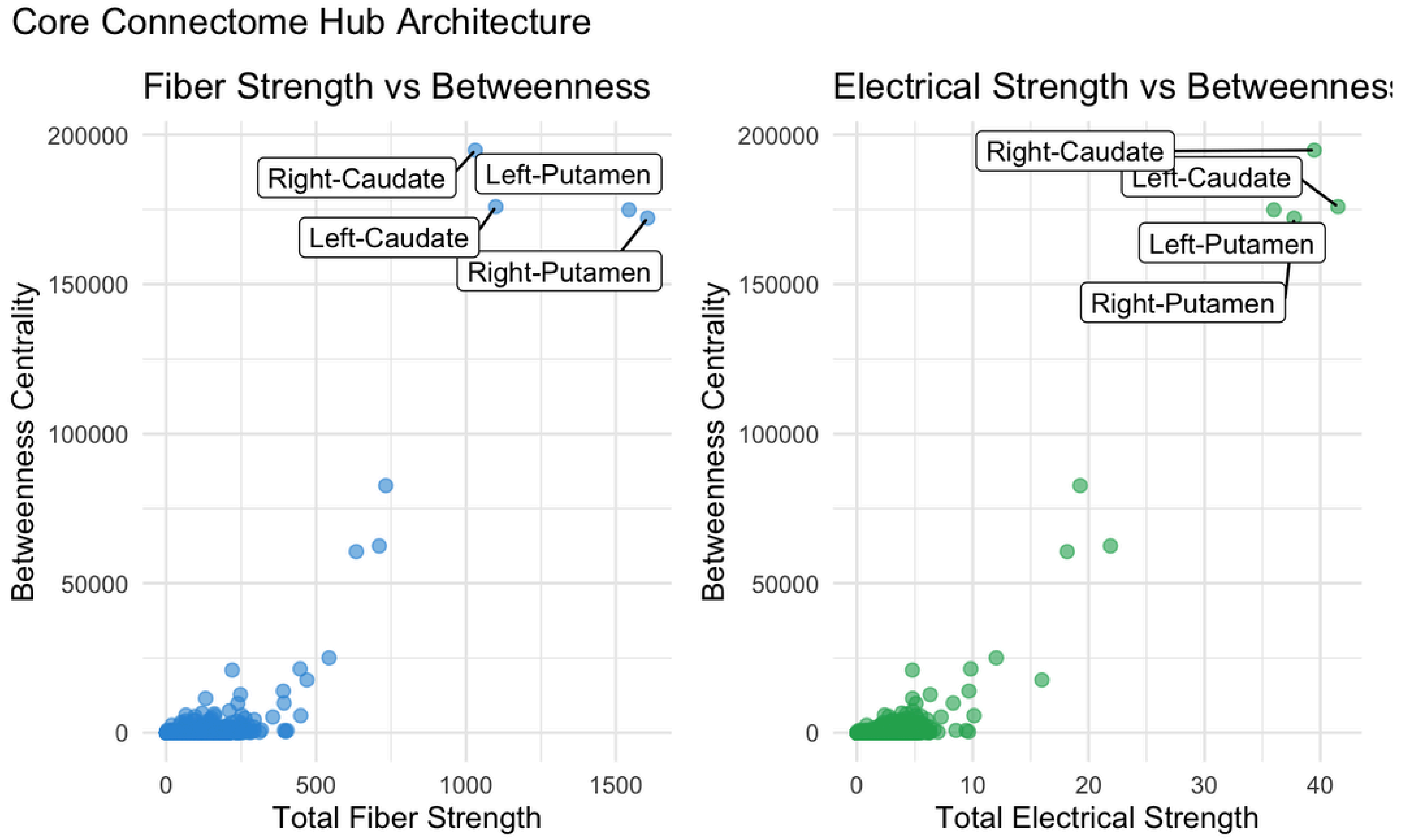



EFFICIENCY RATIO FOR TOP 10 STRUCTUREAL HUBS

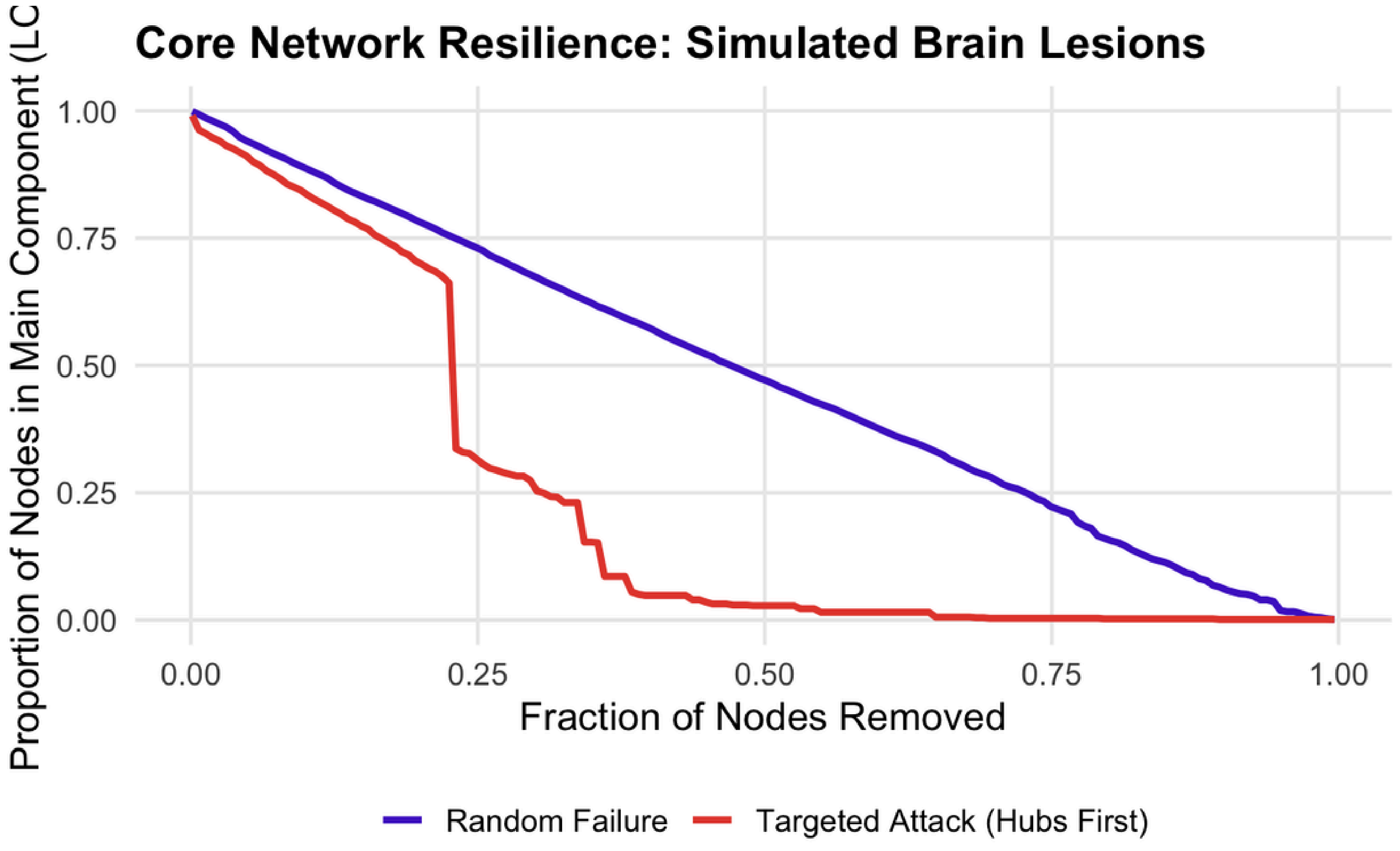
```
# A tibble: 10 × 5
   Node_ID Anatomical_Name          Strength_Fiber Strength_Electrical elec_per_fiber
     <int> <chr>                             <dbl>               <dbl>          <dbl>
 1     502 Right-Caudate                     1031.                39.5         0.0383
 2    1008 Left-Caudate                      1099.                41.5         0.0378
 3     501 Right-Thalamus-Proper              469.                16.0         0.0340
 4    1007 Left-Thalamus-Proper               710.                21.9         0.0308
 5    1012 Left-Hippocampus                   634.                18.1         0.0286
 6     506 Right-Hippocampus                  732.                19.3         0.0263
 7     503 Right-Putamen                     1606.                37.7         0.0235
 8    1009 Left-Putamen                      1544.                36.0         0.0233
 9     631 lh.caudalmiddlefrontal_13          448.                10.1         0.0225
10     158 rh.precentral_7                    543.                12.0         0.0222
```

# 5. BETWEENNESS CENTRALITY



# 5.1 SIMULATION




To test the structural vulnerability of the brain, we systematically removed nodes and tracked the Largest Connected Component.

- Random Failure (Normal Aging): Nodes are removed randomly.
- Targeted Attack (Neurodegeneration): Highest degree hubs are removed first.

The Clinical Reality (Alzheimer's & Dementia): This collapse perfectly mirrors the progression of neurodegenerative diseases.

# 6. COMMUNITIES DETECTION

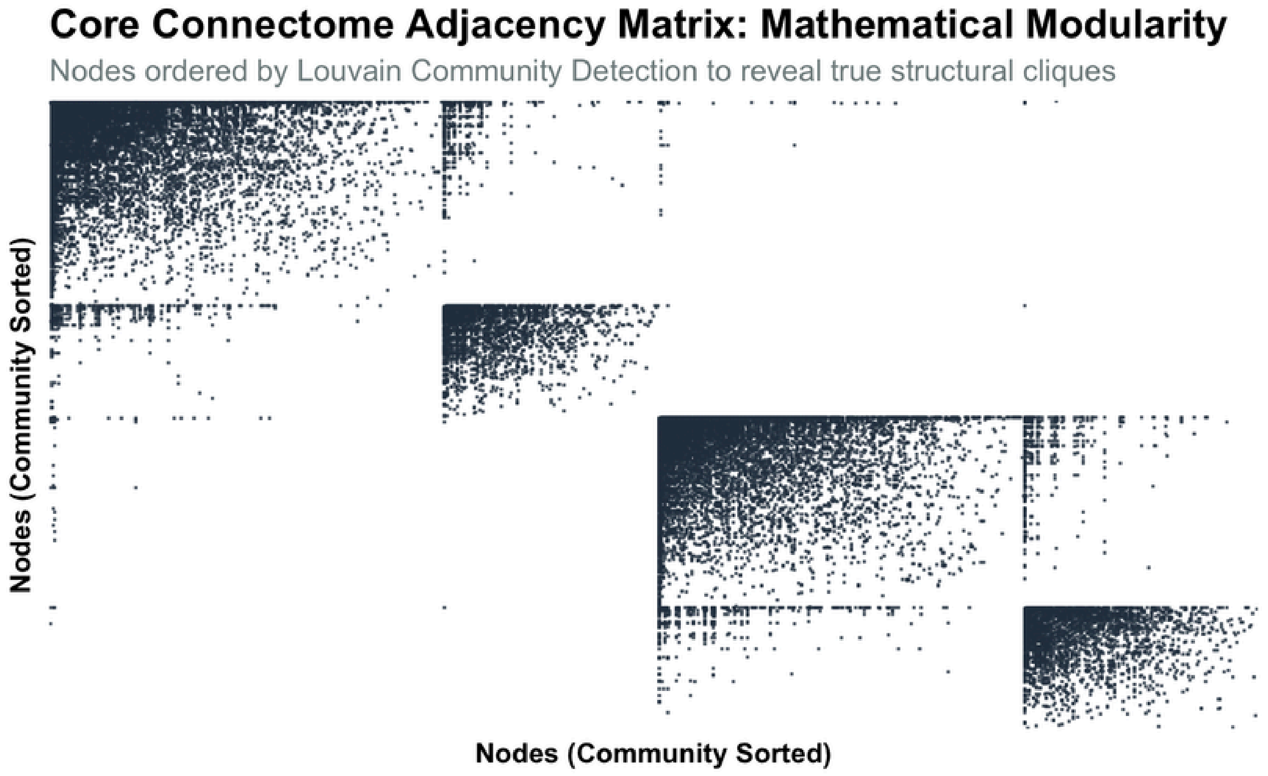
Brain → specialized neighborhoods for task handling

**Louvain Algorithm**
Identifies modules with dense internal wiring and sparse external connections → confirmed by the Matrix Visualization.

Anatomical Mapping
- Communities 1 & 3 – Frontal/central regions → planning, decision-making, motor control
- Communities 2 & 4 – Parietal/occipital/temporal → sensory integration, language, vision

Hemispheric split: algorithm naturally divided the network along the midline

Ratios > 1.0: no side effects of region's large global size detected




| Community | base_fsname | Community_Percentage | Global_Percentage | Ratio |
|:---|:---|---:|---:|---:|
| 1 | superiorfrontal | 14.29 | 9.66 | 1.48 |
| 1 | precentral | 11.97 | 7.30 | 1.64 |
| 1 | rostralmiddlefrontal | 9.27 | 5.77 | 1.61 |
| 1 | postcentral | 7.72 | 4.59 | 1.68 |
| 1 | superiorparietal | 7.34 | 6.48 | 1.13 |
| 2 | inferiorparietal | 14.20 | 4.95 | 2.87 |
| 2 | lateraloccipital | 12.96 | 4.59 | 2.82 |
| 2 | superiortemporal | 9.26 | 3.18 | 2.91 |
| 2 | fusiform | 8.64 | 3.65 | 2.37 |
| 2 | inferiortemporal | 8.64 | 3.18 | 2.72 |
| 3 | superiorfrontal | 16.25 | 9.66 | 1.68 |
| 3 | precentral | 10.83 | 7.30 | 1.48 |
| 3 | rostralmiddlefrontal | 9.03 | 5.77 | 1.56 |
| 3 | superiorparietal | 7.58 | 6.48 | 1.17 |
| 3 | postcentral | 6.86 | 4.59 | 1.49 |
| 4 | inferiorparietal | 12.58 | 4.95 | 2.54 |
| 4 | lateraloccipital | 11.92 | 4.59 | 2.60 |
| 4 | fusiform | 11.26 | 3.65 | 3.08 |
| 4 | inferiortemporal | 8.61 | 3.18 | 2.71 |
| 4 | superiortemporal | 7.95 | 3.18 | 2.50 |

# 7. ERGM MODEL FIT

### Model 1: Baseline Network Density

- The baseline log-odds of a connection is -3.57. This equals a baseline connection probability of just 2.72%.

```
Call:
ergm(formula = g_network ~ edges)

Maximum Likelihood Results:

      Estimate Std. Error MCMC % z value Pr(>|z|)
edges -3.57469    0.01024      0  -349.3   <1e-04 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

     Null Deviance: 499033  on 359976  degrees of freedom
 Residual Deviance:  90055  on 359975  degrees of freedom

AIC: 90057  BIC: 90068  (Smaller is better. MC Std. Err. = 0)
```

### Model 2: Brain Asymmetry and Lateralization

- *nodecov.coord_X* (-0.90): As we move rightward across the brain, the propensity to wire decreases linearly.
- *nodematch.dn_hemisphere* (+5.41): positive coefficient for hemispheric homophily.

```
Call:
ergm(formula = g_network ~ edges + nodecov("coord_X") + nodefactor("dn_hemisphere") +
    nodematch("dn_hemisphere"))

Maximum Likelihood Results:

                               Estimate Std. Error MCMC % z value Pr(>|z|)
edges                          -9.33406    0.18057      0 -51.692   <1e-04 ***
nodecov.coord_X                -0.90299    0.08612      0 -10.485   <1e-04 ***
nodefactor.dn_hemisphere.right -0.15173    0.01618      0  -9.375   <1e-04 ***
nodematch.dn_hemisphere         5.41610    0.14772      0  36.665   <1e-04 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

     Null Deviance: 499033  on 359976  degrees of freedom
 Residual Deviance:  76632  on 359972  degrees of freedom

AIC: 76640  BIC: 76683  (Smaller is better. MC Std. Err. = 0)
```

# 7. ERGM MODEL FIT

**Model 3: Comprehensive Axes Integration**

- *nodefactor.dn_region.subcortical* (+4.60): Subcortical structures (thalamus, hippocampus) exhibit huge connectivity.
- *nodematch.dn_region* (+2.43): Strong homophily based on tissue type.

Controlling for subcortical hubs affects the x-axis coefficient (from -0.90 to -1.77)

$$\text{log-odds}(A_{ij}) = \alpha + (x_i + x_j)\beta + \sum_{k\neq 1}(f_{ik} + f_{jk})\gamma_k + \sum_{h\neq 1}(g_{ih} + g_{jh})\psi_h + I(m_i = m_j)\delta_1 + I(r_i = r_j)\delta_2$$

```
Call:
ergm(formula = g_network ~ edges + nodecov("coord_X") + nodefactor("dn_hemisphere") +
    nodematch("dn_hemisphere") + nodefactor("dn_region") + nodematch("dn_region"))

Maximum Likelihood Results:

                              Estimate Std. Error MCMC % z value Pr(>|z|)
edges                        -13.77423    0.36227      0  -38.02   <1e-04 ***
nodecov.coord_X               -1.77426    0.08657      0  -20.50   <1e-04 ***
nodefactor.dn_hemisphere.right -0.28467   0.01679      0  -16.95   <1e-04 ***
nodematch.dn_hemisphere        6.20265    0.21347      0   29.06   <1e-04 ***
nodefactor.dn_region.subcortical 4.60273  0.17096      0   26.92   <1e-04 ***
nodematch.dn_region            2.43320    0.17297      0   14.07   <1e-04 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

     Null Deviance: 499033  on 359976  degrees of freedom
 Residual Deviance:  72414  on 359970  degrees of freedom

AIC: 72426  BIC: 72491  (Smaller is better. MC Std. Err. = 0)
```

# 7. MODEL VALIDATION

### Statistical Validation (ANOVA)

```
--- TEST 1: LATERAL SPATIAL AXIS SIGNIFICANCE ---
Analysis of Deviance Table

Model 1: g_network ~ edges
Model 2: g_network ~ edges + nodecov("coord_X") + nodefactor("dn_hemisphere") +
    nodematch("dn_hemisphere")
     Df Deviance Resid. Df Resid. Dev Pr(>|Chisq|)
NULL                 359976     499033
1     1   408977     359975      90055    < 2.2e-16 ***
2     3    13423     359972      76632    < 2.2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Test 1: Model 1 (baseline) vs Model 2 (lateralization)
Statistical evidence for Model 2

```
--- TEST 2: DEPTH AXIS ADDED VALUE ---
Analysis of Deviance Table

Model 1: g_network ~ edges + nodecov("coord_X") + nodefactor("dn_hemisphere") +
    nodematch("dn_hemisphere")
Model 2: g_network ~ edges + nodecov("coord_X") + nodefactor("dn_hemisphere") +
    nodematch("dn_hemisphere") + nodefactor("dn_region") + nodematch("dn_region")
     Df Deviance Resid. Df Resid. Dev Pr(>|Chisq|)
NULL                 359976     499033
1     4   422401     359972      76632    < 2.2e-16 ***
2     2     4218     359970      72414    < 2.2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Test 2: Model 2 (lateralization) vs Model 3 (depth)
Statistical evidence for Model 3

# THANK YOU!